

The Best Open Source BLOCKCHAIN Platforms

This article presents the author's choice of the best open source blockchain platforms. Of course, we realise that opinions will always differ on what is the 'best'. But to have an argument, there must be a starting point. Perhaps readers might like to share their preferences, if they happen to be different from what is featured here.



A blockchain is basically a time-stamped series of immutable records of data that is managed by a cluster of computers and not by any single entity. Each block of data is secured and bound to the other using various cryptographic principles, thus forming a chain. There is no central authority for the blockchain network. It's more like a democratised system with a shared and immutable ledger, with the information open for everyone to see. So we can say that anything built using blockchain technology is very transparent by its very nature, and everyone involved in it is accountable for his or her actions.

Each blockchain is a linked list that contains data and a hash pointer (a pointer containing the hash of the data as well as the address of the previous block) pointing to its previous block, and hence generating the whole blockchain. Let's try to understand this with an example. When we book railway tickets through any mobile app or the Web, the credit or debit card company takes its commission for processing the transaction. If we consider migrating the entire ticketing system to blockchain technology, then we need to consider only the two parties in the transaction – the railway company and the passenger. Each booked ticket is a block that is continuously added to the complete ticket blockchain. Just as every monetary transaction on the blockchain is a unique,

tamper-proof and independently verifiable record, so is the case with each of the booked tickets. Hence, the final ticket blockchain is nothing but a record containing all transactions for, say, the entire train network or a certain train journey, comprising every ticket sold.

The most important part here is that all of this includes just the blockchain setup cost and the rest is free. Not only can the blockchain technology help us transfer and store money, it can also replace all the processes and various business models that rely on charging a small transaction fee. The blockchain is really a simple yet ingenious way of passing different information from A to B in an automated and safe manner. Falsifying any single record would mean falsifying the complete chain in millions of instances, which is virtually impossible. Bitcoin, the pioneering blockchain based cryptocurrency, uses the same model for various monetary transactions.

The pillars of blockchain technology

Here are the three significant properties of a blockchain that have made it a preferred technology in the market.

Decentralisation: Before Bitcoin came into existence, we were more used to a model in which we solely interacted with a centralised entity that stored all the data, in order to extract whatever information we required. As we all know, such